

4COSC002W MATHEMATICS FOR COMPUTING

Week 7 Seminar Tasks

Graph Theory

READING

Lecture 5 Notes (available on Blackboard)

Chapters 10-11. Grossman, P., 2017. *Discrete mathematics for computing*. Bloomsbury Publishing.

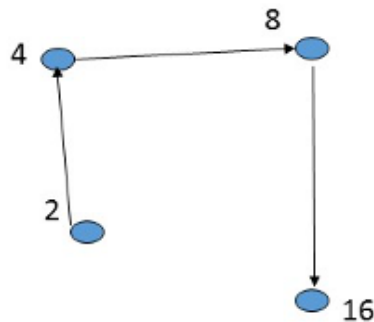
TASK 1. GRAPH NOTATION. GRAPH TYPES

See Lectures 5 Notes -Slides 5-10

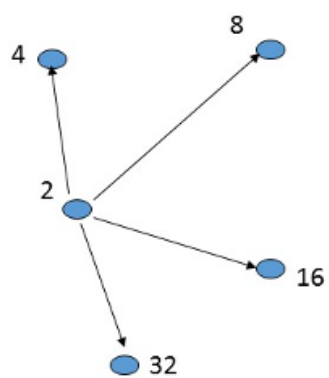
Task 1.1. For each of the following graphs represented in the diagrams below do the following:

- form set V of all its nodes and set E of all its edges
- determine if it is a directed or undirected graph
- determine if it is a cyclic or acyclic graph, for a cyclic graph give an example of a cyclic path in it
- is there a pattern a graph can represent?

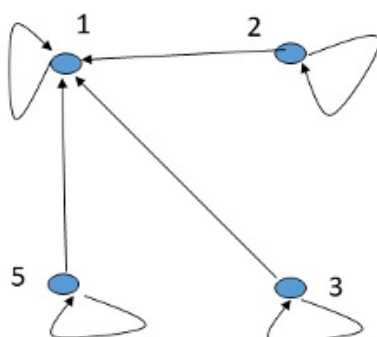
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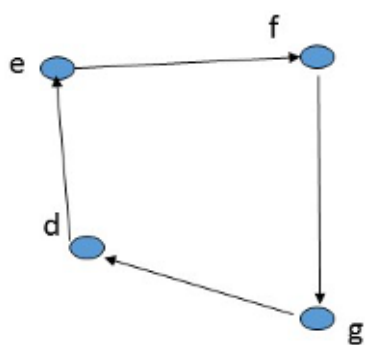
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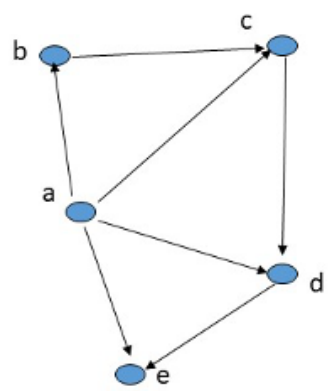
c.



d.



e.



Task 1.2. For each of the following graphs defined by the given set of nodes, V , and set of edges, E , do the following:

- draw its representation corresponding to the definition of V and E
- determine if it is a directed or undirected graph
- determine if it is a cyclic or acyclic graph, for a cyclic graph give an example of a cyclic path in it.
- calculate the sum of the degrees of all vertices

A. $V=\{d,e,f,g\}$; $E=\{\{d,e\}, \{e,f\}, \{f,g\}, \{g,d\}\}$

B. $V=\{a,b,c,d,e\}$; $E=\{\{a,b\}, \{a,c\}, \{a,d\}, \{a,e\}, \{b,c\}, \{c,d\}, \{d,e\}\}$

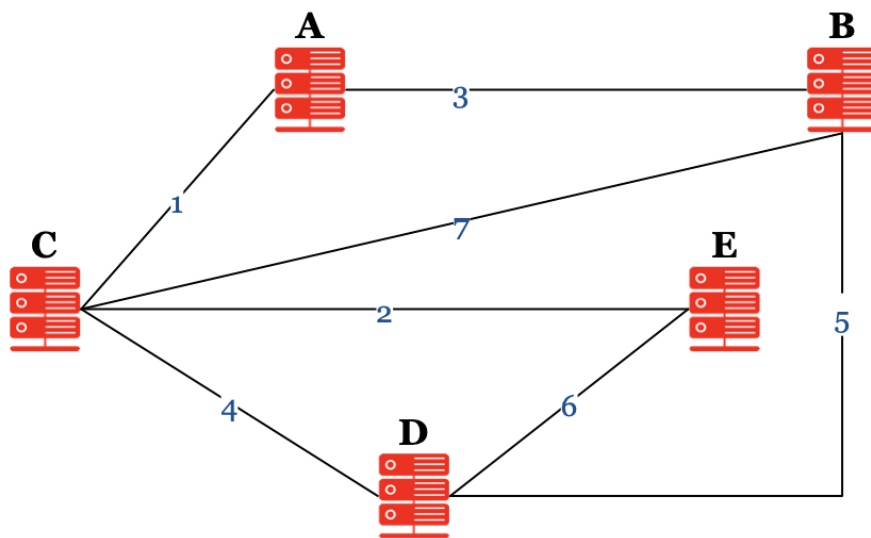
C. $V=\{a,b,c\}$; $E=\{(a,a), (a, b), (b,b), (b,a), (b, c), (c,c), (c,b), (a,c), (c,a)\}$

TASK 2. WEIGHTED GRAPHS

See Lectures 5 Notes -Slide 11-14

In a distributed database system, several servers (nodes) store parts of a database and are connected via direct data transfer links (edges) with specific costs, representing the time or resources needed to transfer data between servers.

Weighted Graph:



Servers (Nodes): A, B, C, D, E

Connections (Edges) and Transfer Costs (in milliseconds):

- A-B: 3
- A-C: 1
- B-C: 7
- B-D: 5
- C-D: 4
- C-E: 2
- D-E: 6

Tasks:

1. Identify Key Connections for Fastest Transfers:
 - a. Which connections have the lowest and highest data transfer costs?
 - b. How do these connections impact overall transfer efficiency?
2. Calculate Average Transfer Cost Across the Network:
 - a. What is the average data transfer cost between servers in the system?
 - b. Interpret what this average means in terms of overall system efficiency.
3. Find Alternative Paths for Reliability:

- a. If the connection between B and C becomes unreliable, list alternative paths for data transfer between B and C, and calculate the total transfer costs for each path.
 - b. Which alternative path is the most efficient?
4. Data Balancing Optimisation:
 - a. Assuming that servers can occasionally be reconfigured to reduce transfer costs, analyse which server-to-server connections could be optimised (i.e., connections with the highest costs that could be improved).
 - b. Suggest a way to distribute data or reorganise connections to improve data transfer efficiency without losing the direct connectivity between all servers.
5. Determine Server Load Based on Node Degree:
 - a. Calculate the degree of each server in the network.
 - b. Based on the degree, identify which server(s) could be experiencing the highest load due to a large number of direct data transfers.
 - c. Provide suggestions for balancing the load by analysing the degree of connectivity.

TASK 3. TREES.

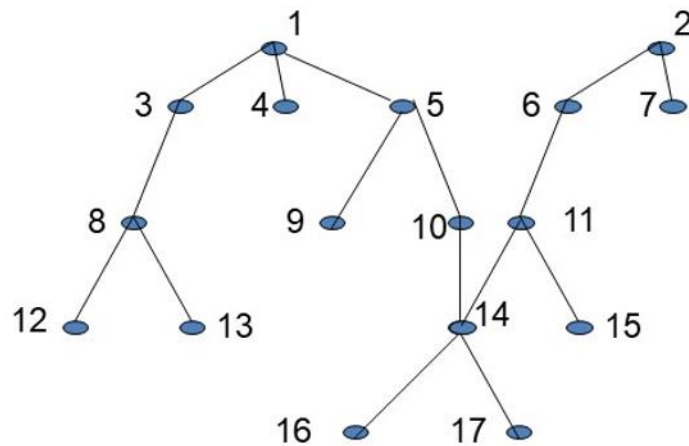
See Lectures 5 Notes -Slides 24-29

In the figures below, you will see attempts to draw rooted trees. Justify if the graphs drawn are trees.

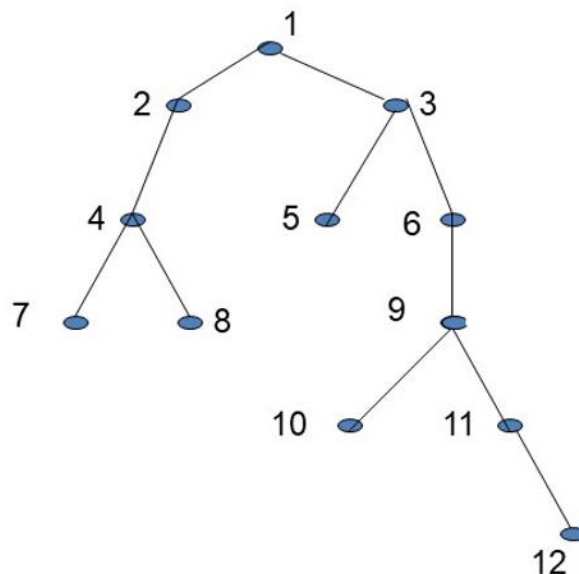
For those cases where you established trees, identify:

- All the leaves of a tree
- Depth of a tree
- If a tree is a binary tree

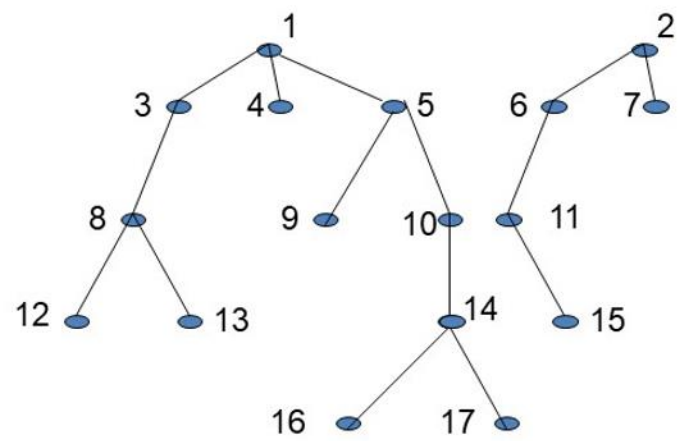
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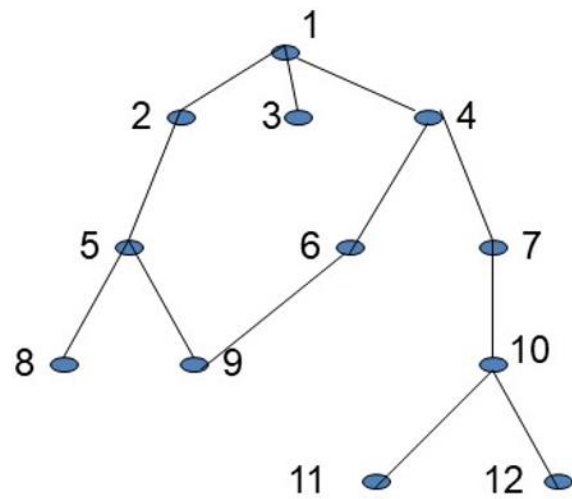
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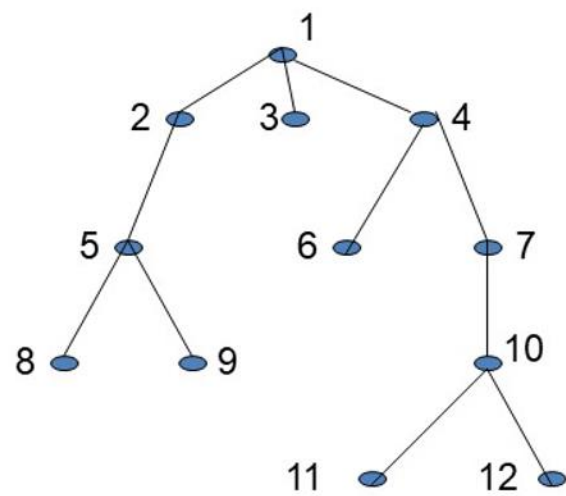
c.



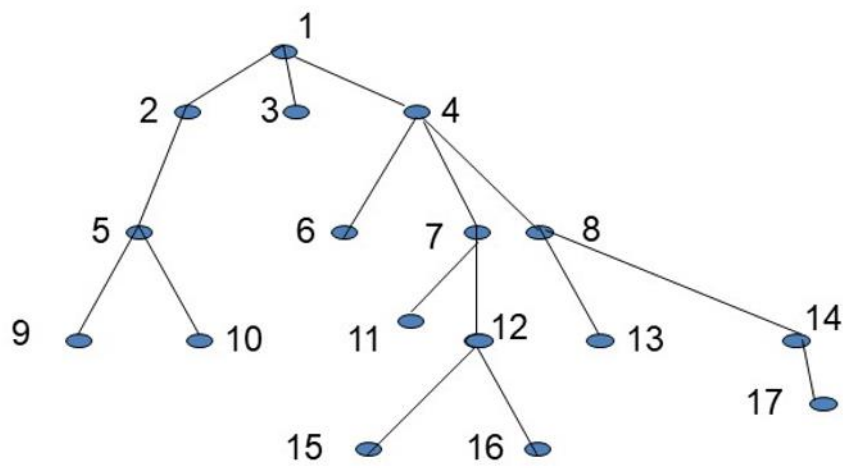
d.



e.



f.



TASK 4: CRITICAL PATH ANALYSIS IN PROJECT SCHEDULING

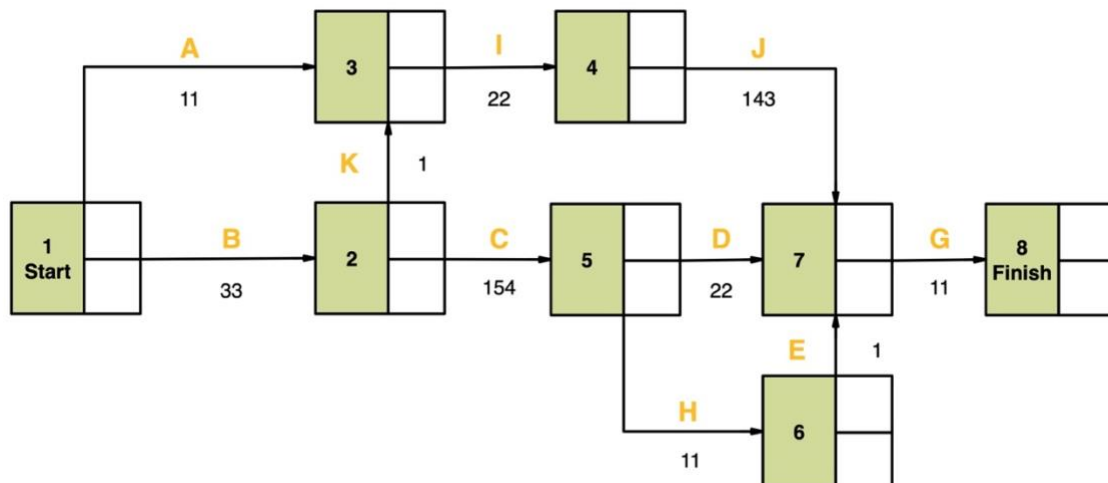
See Lectures 5 Notes -Slides 15-22

Task A

The provided diagram represents a project schedule using a network graph, where each node (or box) represents a task within the project. The edges represent activity and are labelled with letters, and the duration of each task is given below. The arrows indicate the sequence of tasks, demonstrating which tasks must be completed before others can begin.

Activity	Description	Start Point	End Point	Duration (Days)
A	Requirement Analysis	1	3	11
B	Database Setup	1	2	33
C	Data Integration	2	5	154
D	Testing Module	5	7	22
E	Deployment Preparation	6	7	1
G	System Testing	7	8	11
H	User Interface Design	5	6	11
I	Security Protocol Design	3	4	22
J	Load Balancing Setup	4	7	143
K	API Development	3	2	1

Network Graph:



Using the provided network graph, your objective is to perform a critical path analysis. This will involve identifying the longest path through the network, which determines the shortest time in which the project can be completed.

NB: The **critical path** is the sequence of stages with *no slack time*; if any of these stages are delayed, the entire project will be delayed.

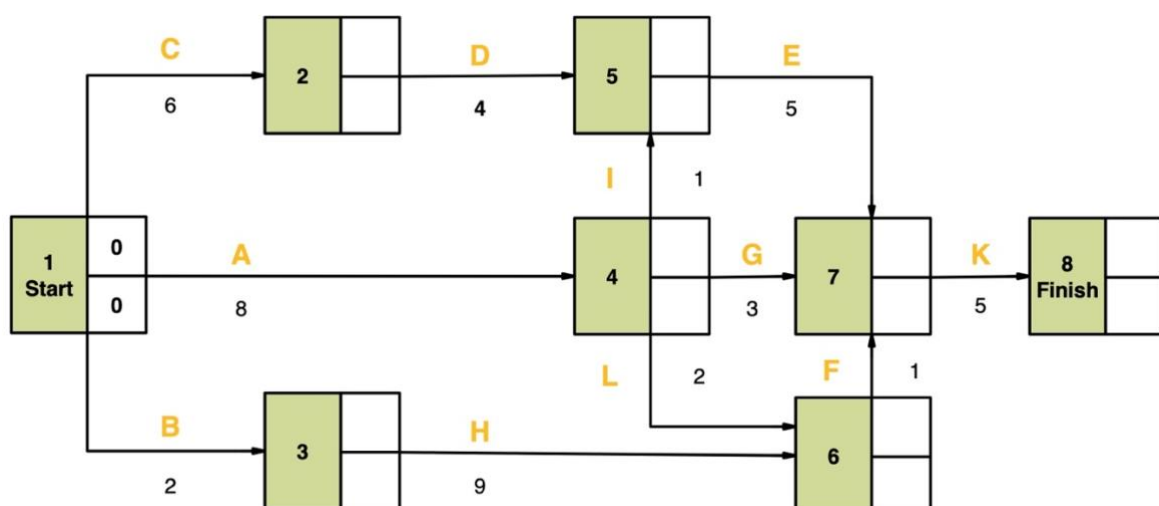
For your analysis, follow these steps:

- Calculate the Earliest Start Time (EST) and Latest Start Time (LST) for each task.
- Determine the Total Float (or Slack) for each task, which is the difference between the LST and EST.
- Identify the critical path, which is the path from Start to Finish that has zero slack for all tasks.

Task B

The provided diagram represents a project schedule using a network graph, where each node (or box) represents a task within the project. The edges represent activity and are labelled with letters, and the duration of each task is given below. The arrows indicate the sequence of tasks, demonstrating which tasks must be completed before others can begin.

Network Graph:



Using the provided network graph, your objective is to perform a critical path analysis. This will involve identifying the longest path through the network and determining when the project can be completed.

For your analysis, follow these steps:

- A. Fill in all the empty cells (EST and LFT) for each node.
- B. Provide detailed calculations of the Earliest Start Time (EST, the upper empty cell of the node) and the Latest Finish Time (LFT, the lower empty cell of the node) for tasks 4 and 6 only.
- C. Identify the critical path, which is the path from Start to Finish that has zero slack for all tasks. Show the critical path on your drawn graph.